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सत्यमेव जयते



पेटेंट कार्यालय, भारत सरकार

The Patent Office, Government Of India

पेटेंट प्रमाण पत्र

Patent Certificate

(पेटेंट नियमावली का नियम 74)

(Rule 74 of The Patents Rules)

पेटेंट सं. / Patent No.

504302

आवेदन सं. / Application No.

201931018283

फाइल करने की तारीख / Date of Filing

07/05/2019

पेटेंटी / Patentee

INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR

प्रमाणित किया जाता है कि पेटेंटी को, उपरोक्त आवेदन में यथाप्रकरित *A BUCK CONVERTER SYSTEM TO MITIGATE TRANSIENTS* नामक आविष्कार के लिए, पेटेंट अधिनियम, 1970 के उपबंधों के अनुसार आज तारीख मई 2019 के सातवें दिन से बीस वर्ष की अवधि के लिए पेटेंट अनुदत्त किया गया है।

It is hereby certified that a patent has been granted to the patentee for an invention entitled *A BUCK CONVERTER SYSTEM TO MITIGATE TRANSIENTS* as disclosed in the above mentioned application for the term of 20 years from the 7th day of May 2019 in accordance with the provisions of the Patents Act, 1970.



[Signature]

Controller of Patents

अनुदान की तारीख : 29/01/2024
Date of Grant :

टिप्पणी - इस पेटेंट के नवीकरण के लिए फीस, यदि इसे बनाए रखा जाना है, मई 2021 के सातवें दिन को और उसके पश्चात प्रत्येक वर्ष में उसी दिन देय होगी।

Note. - The fees for renewal of this patent, if it is to be maintained, will fall / has fallen due on 7th day of May 2021 and on the same day

in every year thereafter.

FORM 1 THE PATENTS ACT 1970 (39 of 1970) and THE PATENTS RULES, 2003 APPLICATION FOR GRANT OF PATENT (See section 7, 54 and 135 and sub-rule (1) of rule 20)				(FOR OFFICE USE ONLY)	
				Application No.	
				Filing date:	
				Amount of Fee paid:	
				CBR No:	
				Signature:	
1. APPLICANT'S REFERENCE / IDENTIFICATION NO. (AS ALLOTTED BY OFFICE)					
2. TYPE OF APPLICATION *Please tick (✓) at the appropriate category+					
Ordinary (✓)		Convention ()		PCT-NP ()	
Divisional ()	Patent of Addition ()	Divisional ()	Patent of Addition ()	Divisional ()	Patent of Addition ()
3A. APPLICANT(S)					
Name in Full		Nationality	Country of Residence	Address of the Applicant	
INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR		An Indian Statutory Body.	INDIA	House No.	Sponsored Research & Industrial Consultancy, Indian Institute of Technology, Kharagpur
				Street	
				City	Kharagpur
				State	West Bengal
				Country	India
				Pin code	721302
3B. CATEGORY OF APPLICANT *Please tick (✓) at the appropriate category+					
Natural Person ()		Other than Natural Person			
		Small Entity	Startup ()	Others (✓)	
4. INVENTOR(S) *Please tick (✓) at the appropriate category+					

Are all the inventor(s) same as the applicant(s) named above?	Yes ()	No (✓)		
If “No”, furnish the details of the inventor(s)				
Name in Full	Nationality	Country of Residence		
ACHARYA, Arnab;	Indian	India	House No.	M.S Research scholar, Department of Electrical Engineering, Indian Institute of Technology, Kharagpur;
			Street	
			City	Kharagpur
			State	West Bengal
			Country	India
			Pin code	721302
KUMAR, V Inder;	Indian	India	House No.	SRF, Department of Electrical Engineering, Indian Institute of Technology, Kharagpur;
			Street	
			City	Kharagpur
			State	West Bengal
			Country	India
			Pin code	721302
KAPAT, Dr. Santanu;	Indian	India	House No.	Associate Professor, Department of Electrical Engineering, Indian Institute of Technology, Kharagpur;
			Street	
			City	Kharagpur
			State	West Bengal
			Country	India
			Pin code	721302
5. TITLE OF THE INVENTION:				

A BUCK CONVERTER SYSTEM TO MITIGATE TRANSIENTS.					
6. AUTHORISED REGISTERED PATENT AGENT(S)			IN/PA No.		199
			Name		ANJAN SEN
			Mobile No.		9830050839
7. ADDRESS FOR SERVICE OF APPLICANT IN INDIA			Name		ANJAN SEN & ASSOCIATES
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			Telephone No.		033-24749871;033- 65293955
			Mobile No.		9830050839
			Fax No.		033-24868693
			E-mail ID		info@ipindiaasa.com
8. IN CASE OF APPLICATION CLAIMING PRIORITY OF APPLICATION FILED IN CONVENTION COUNTRY, PARTICULARS OF CONVENTION APPLICATION					
Country	Application Number	Filing date			IPC (as classified in the convention country)
9. IN CASE OF PCT NATIONAL PHASE APPLICATION, PARTICULARS OF INTERNATIONAL APPLICATION FILED UNDER PATENT CO-OPERATION TREATY (PCT)					
International application number			International filing date		

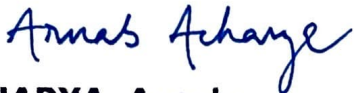
10. IN CASE OF DIVISIONAL APPLICATION FILED UNDER SECTION 16, PARTICULARS OF ORIGINAL (FIRST) APPLICATION	
Original (first) application No.	Date of filing of original (first) application
11. IN CASE OF PATENT OF ADDITION FILED UNDER SECTION 54, PARTICULARS OF MAIN APPLICATION OR PATENT	
Main application/patent No.	Date of filing of main application
12. DECLARATIONS	

(i) Declaration by the inventor(s)

(In case the applicant is an assignee: the inventor(s) may sign herein below or the applicant may upload the assignment or enclose the assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period).

We, the above named inventor(s) ~~is~~/are the true & first inventor(s) for this Invention and declare that the applicant(s) herein ~~is~~/are ~~my~~/our assignee or legal representative.

(a) Date: 07.05.2019

(b) Signature(s): 

(c) Name(s) : **ACHARYA, Arnab;**

(a) Date: 07.05.2019

(b) Signature(s): 

(c) Name(s) : **KUMAR, V Inder;**

(a) Date: 07.05.2019

(b) Signature(s): 

(c) Name(s) : **KAPAT, Dr. Santanu;**

(ii) ~~Declaration by the applicant(s) in the convention country~~

~~(In case the applicant in India is different than the applicant in the convention country: the applicant in the convention country may sign herein below or applicant in India may upload the assignment from the applicant in the convention country or enclose the said assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period)~~

~~I/We, the applicant(s) in the convention country declare that the applicant(s) herein ~~is~~/are ~~my~~/our assignee or legal representative.~~

~~(a) Date~~

~~(b) Signature(s)~~

~~(c) Name(s) of the signatory~~

(iii) Declaration by the applicant(s)

I/We the applicant(s) hereby declare(s) that: -

- ☒ We are in possession of the above-mentioned invention.
- ☒ The ~~provisional~~/complete specification relating to the invention is filed with this application.
- ☒ ~~The invention as disclosed in the specification uses the biological material from India and the necessary permission from the competent authority shall be submitted by me/us before the grant of patent to me/us.~~
- ☒ There is no lawful ground of objection(s) to the grant of the Patent to me/us.
- ☒ ~~We are the true & first inventor(s).~~
- ☒ We are the assignee or legal representative of true & first inventor(s).
- ☒ ~~The application or each of the applications, particulars of which are given in Paragraph 8, was the first application in convention country/countries in respect of my/our invention(s).~~
- ☒ ~~I/We claim the priority from the above mentioned application(s) filed in convention country/countries and state that no application for protection in respect of the invention had been made in a convention country before that date by me/us or by any person from which I/We derive the title.~~
- ☒ ~~My/our application in India is based on international application under Patent Cooperation Treaty (PCT) as mentioned in Paragraph 9.~~
- ☒ ~~The application is divided out of my /our application particulars of which is given in Paragraph 10 and pray that this application may be treated as deemed to have been filed on DD/MM/YYYY under section 16 of the Act.~~
- ☒ ~~The said invention is an improvement in or modification of the invention particulars of which are given in Paragraph 11.~~

13. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION

(a) Form 2

Item	Details	Fee	Remarks
Complete/ provisional specification)#	No. of pages – 12 (including Description)	Rs. 8,000/- on application	
No. of Claim(s)	No. of claims -10 and No. of pages- 3		
Abstract	No. of pages -1		

No. of Drawing(s)	No. of drawings -4 and No. of pages -3		
# In case of a complete specification, if the applicant desires to adopt the drawings filed with his provisional specification as the drawings or part of the drawings for the complete specification under rule 13(4), the number of such pages filed with the provisional specification are required to be mentioned here. (b) Complete specification (in conformation with the international application)/as amended before the International Preliminary Examination Authority (IPEA), as applicable (2 copies). (c) Sequence listing in electronic form (d) Drawings (in conformation with the international application)/as amended before the International Preliminary Examination Authority (IPEA), as applicable (2 copies). (e) Priority document(s) or a request to retrieve the priority document(s) from DAS (Digital Access Service) if the applicant had already requested the office of first filing to make the priority document(s) available to DAS. (f) Translation of priority document/Specification/International Search Report/International Preliminary Report on Patentability. (g) Statement and Undertaking on Form 3 (h) Declaration of Inventorship on Form 5 (i) Power of Authority (j)..... Total Fee of Rs. 8,000/- by online payment through SBI with e-filing. We hereby declare that to the best of our knowledge, information and belief the fact and matters slated herein are correct and we request that a patent may be granted to us for the said invention. Dated this 7th day of May, 2019 Signature: Name: To The Controller of Patents The Patent Office, at Kolkata.			



Anjan Sen
Of Anjan Sen & Associates
(Applicants Agent)
IN/PA-199